



MINING DEALER BEST PRACTICE

D11 Series Pivot Shaft Block Tool

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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November 2013
1306-4.03-1318

1.0 Introduction

During the D11 Track roller frame removal and installation process, it is necessary to use some tools in order to ensure a safe procedure. This process is referenced in the document SENR1782-08 or RENR8226-03 for one frame and it should be applied in the same way for the second one. However, the document does not mention another component that should be blocked during the removal of the two track roller frames which is the pivot shaft.

This Best Practice describes the tool which allows to block the pivot shaft and keep its centered position on the main frame during the removal and installation process of the two track roller frames.

2.0 Best Practice Description

During the two track rollers frames removal and installation process, it is necessary to block the pivot shaft to keep it from moving freely alongside the hole in the main frame. When removing a singer roller frame, it is not necessary to do this because the other track roller frame restricts the pivot shaft movement (See figure 1).

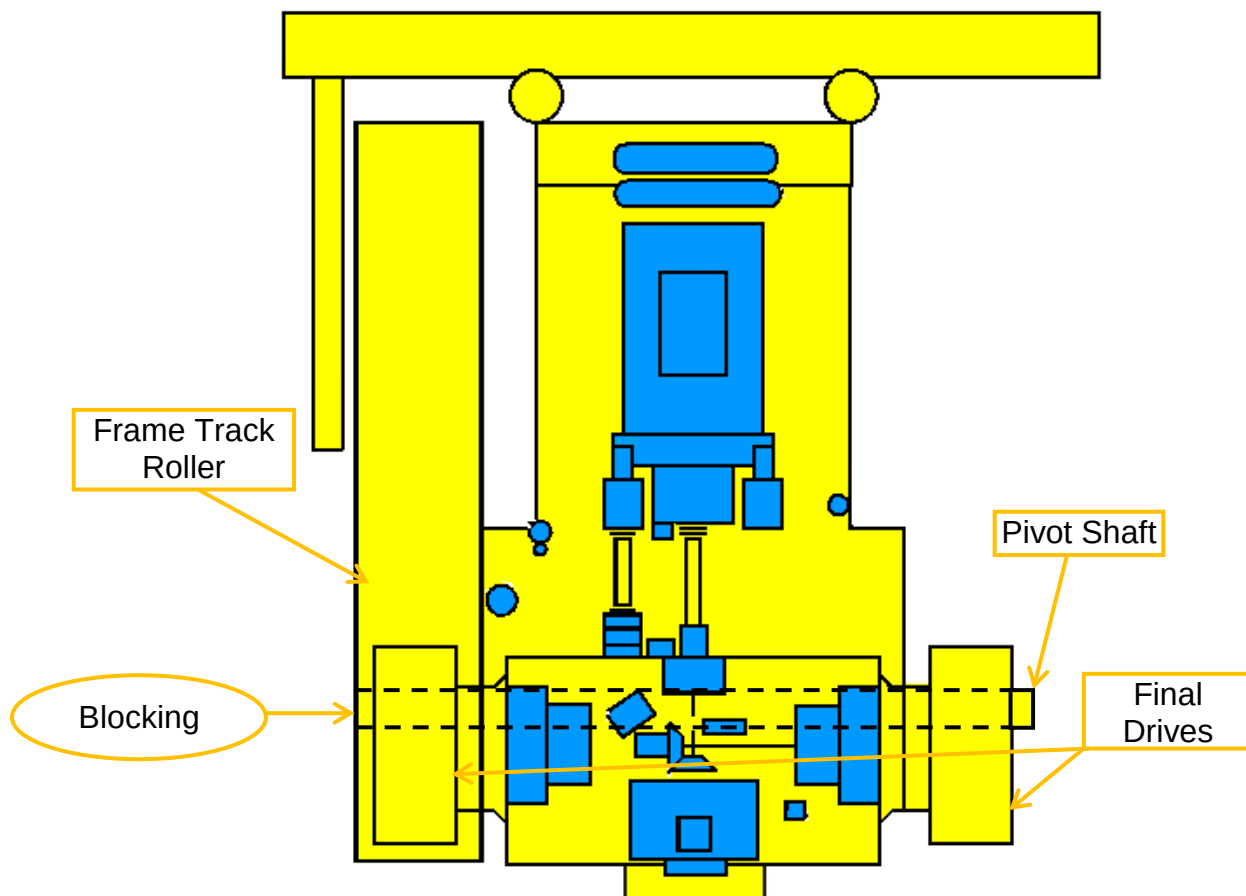


Figure . Pivot shaft blocking by the Track Roller Frame.

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The same process should be applied when a track roller frame is installed while the other one is in place; it is not necessary to block the pivot shaft and the described procedure on SIS Web is applied just the way it shows.

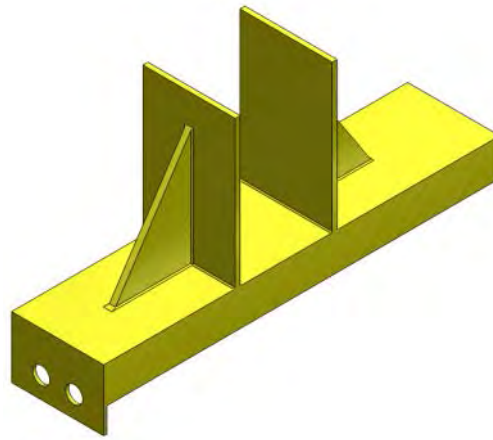


Figure 2. Pivot Shaft block tool

The tool is installed at the opposite side of the roller frame to be removed. The pivot shaft is secured by 2 bolts, using neoprene to protect the component surface. The support plates block the shaft movement resting on the closest final drive (See Figure 2 and 3)

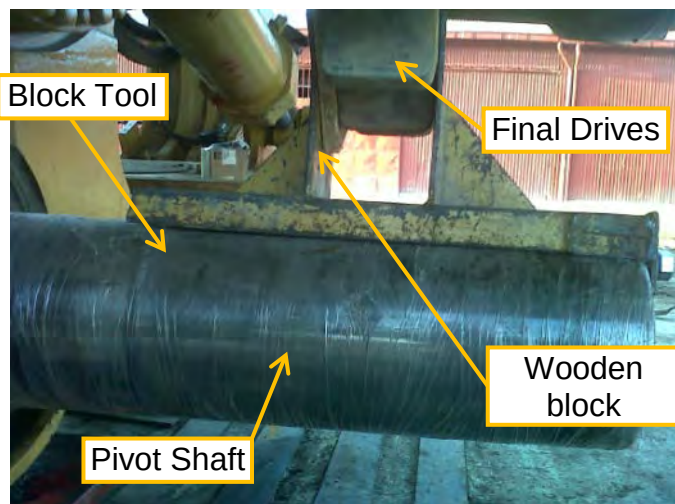


Figure 3. Pivot Shaft blocking with the tool



Figure 4. Tool Location on pivot shaft

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3.0 Implementation Steps

Steps for track roller frame Removal:

1. Refer to disassembly document SENR1782-08 (D11R Track Roller Frame Removal) or RENR8226-03 (D11T Track Roller Frame Removal) and follow the steps to remove one track roller frame.
2. Place neoprene piece on pivot shaft face.
3. Locate the tool securing it with two bolts and washers to the pivot shaft. The bolts used depend on the pivot shaft (see Table 2), which minimum torque is 25 N • m. The closest final drive should be between the two support plates, place a wooden block to adjust it to the tool (See Figure 3 and 4)
4. Once the pivot shaft is secured; continue with the removal of the other track roller frame as indicated on SENR1782-08 or RENR8226-03

Steps for the track roller frames Installation:

Follow the steps above using the installation document SENR1782-08 or RENR8226-03.

The tool can be used for the following pivot shaft (D11N, D11R & D11T Tractors):

Table 1. Pivot Shafts	
Part Number	Series
7T-5713	4HK, 74Z, 8ZR
134-2202	7PZ, 9TR, 9XR, AAF, AMA, GEB, JEL, JNS, MDG, TPB

4.0 Benefits

- Improved safety during pivot shaft removal and installation process to prevent it from going out and cause damage or harm to the technicians.
- It maintains the horizontal position (centered) the component must have.
- It decreases time and component handling complexity.

5.0 Resources Required

- Hardware list (see table 2)
- Neoprene piece (dimensions: 26 x 8 x 1/16 inch)
- Wooden block
- Technical Personnel (at least two)
- Material and equipment required to make the tool

Table 2. Hardware Required		
Part Number	Description	Qty
1D-4625	7/8-9 x 2.50 inch Bolt (for 7T-5713 pivot shaft)	2
1D-4637	1-8 x 3.25 inch Bolt (for 134-2202 pivot shaft)	2
5P-8250	1 inch Washer	2

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6.0 Supporting Attachments / References

- SIS document: SENR1782-08 (Track Roller Frame Remove SMCS: 4151-011)
- SIS document: SENR1782-08 (Track Roller Frame Install SMCS: 4151-012)
- SIS document: RENR8226-03 (Track Roller Frame Remove SMCS: 4151-011)
- SIS document: RENR8226-03 (Track Roller Frame Install SMCS: 4151-012)
- D11 Series Pivot Shaft Tool Plane

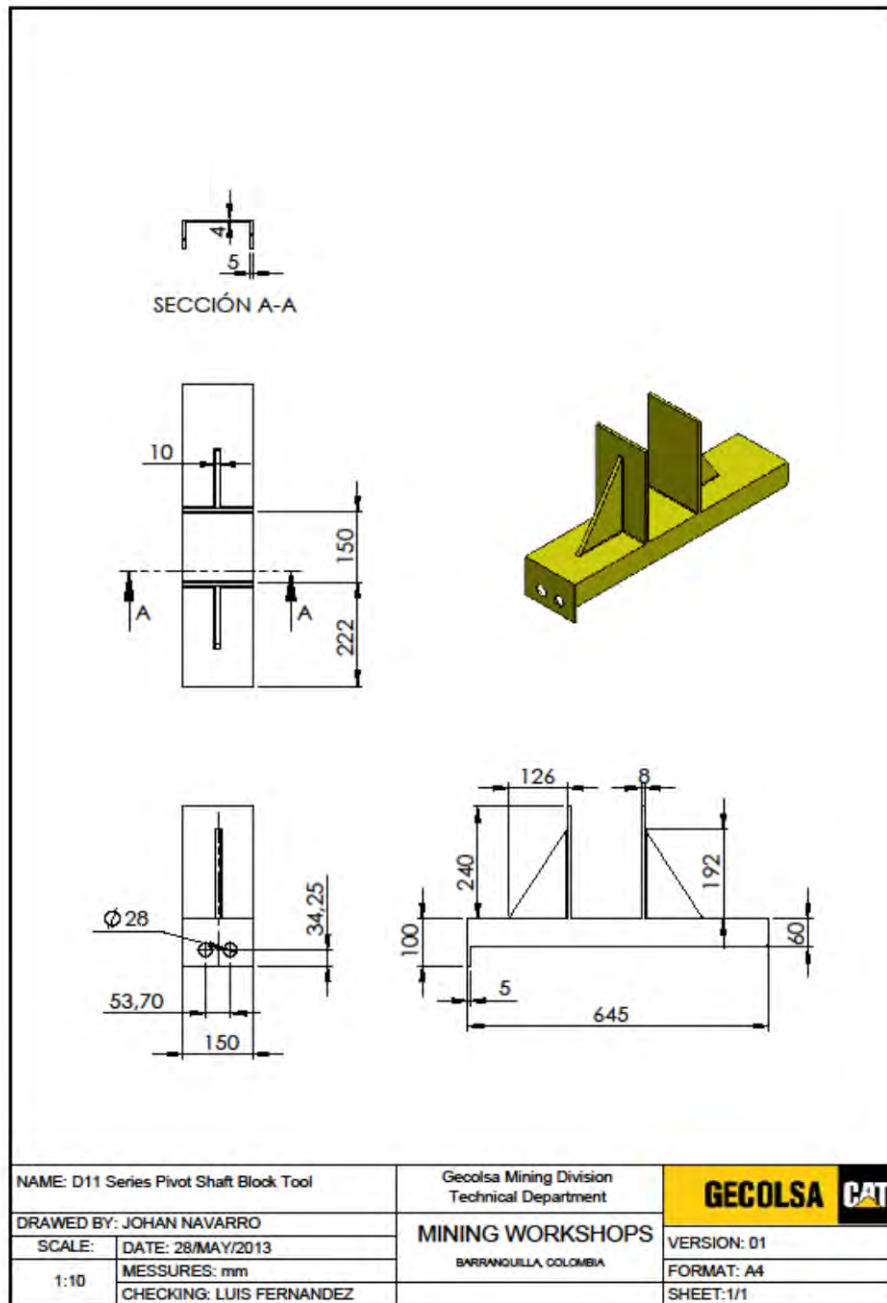


Figure 5. Tool Drawing.

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- Finite Element Analysis for block Tool

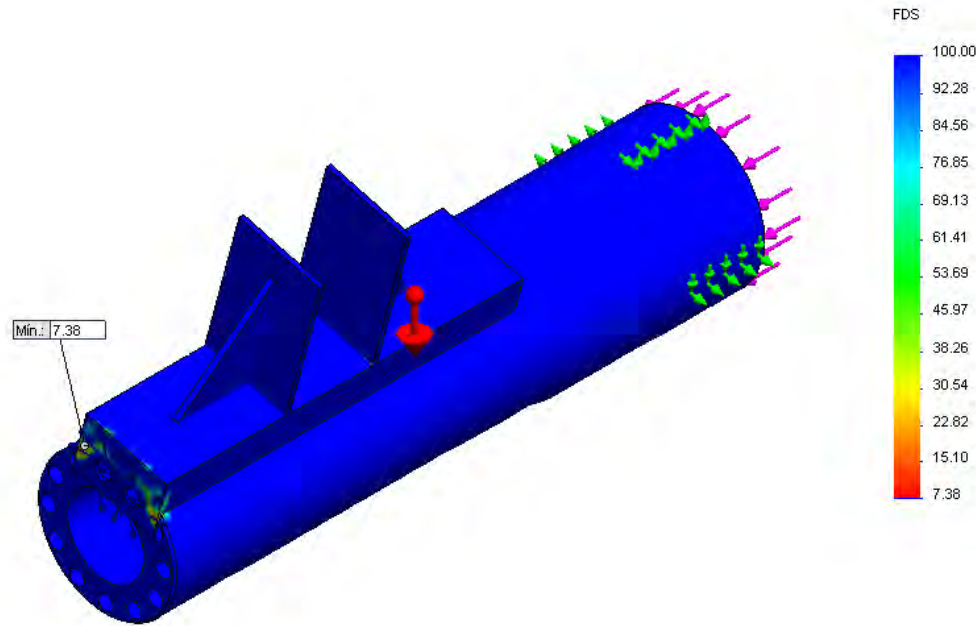


Figure 6. Safety Factor analysis

Physical Properties		
Material	ASTM A36 Steel	
Mass	12.82 Kg	
Density	7850 Kg/m ³	
Yield Strength	250 MPa	
Tensile Strength	400 MPa	
Shear Modulus	79.3 Gpa	
Study Properties		
Analysis type	Static	
Mesh type	Solid mesh	
Mesh Quality	Second order tetrahedral elements	
Model type	Lineal elastic isotropic	
Failure criteria	Von Mises Stress Maximum	
Fixtures		
Type	Fixtures details	
Bolt	Quantity	2
	Nominal Diameter	1 inch
	Pre-load (Torque)	25 N·m
Virtual Wall	Quantity	2
	Entities	Support plate inside faces
	Wall type	Bonded
	Friction coefficient	0.15
Faces Cylindrical	Entities	Frame bearing support
	Restriction	Radial and circumferential

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Loads	
Gravity	9.81 m/s ²
Horizontal Force	1960 N (200 kg.)
Study Results	
Von Mises Maximum Stress	96.1 Mpa
Minimum Safety Factor	7.38

Based on the finite element analysis, the tool is subjected to a thrust force of 200 kg (pivot shaft trying to come out) has a factor of safety 7.38.

7.0 Related Best Practices

8.0 Acknowledgements

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 The Development and Engineering Area Team

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